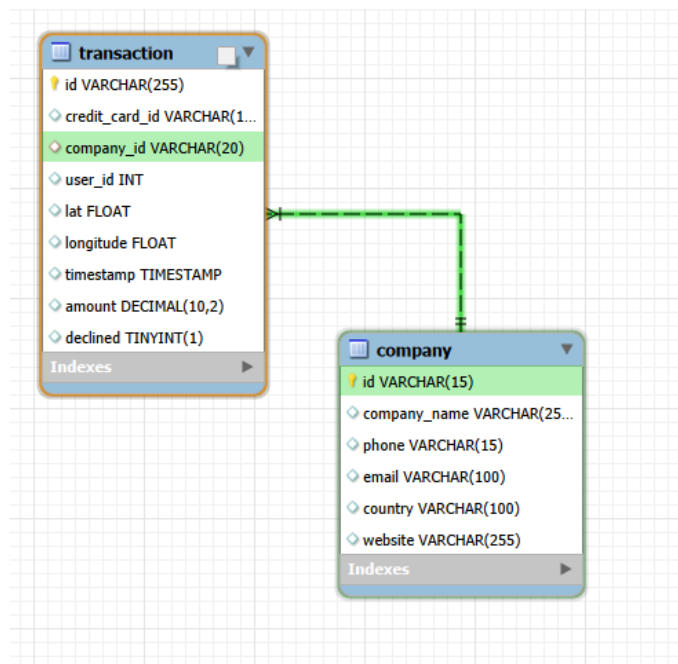


SPRINT 2

NIVELL 1

Exercici 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.



Per fer-ho he carregat els arxius amb *Open SQL Script*, primer *estructura_dades* i després *dades_introduir*.

Es tracta de dues taules amb relació 1 (*company*) a N (*transaction*).

La taula *company* és una taula de dimensió ja que la seva informació es descriptiva i no es pot fer cap tipus d'operació amb les seves dades. La seva **primary key** és *id*.

La taula *transaction* és una taula de fets que té una **foreign key** (*company_id*) que és la **primary key** de *company*, i té una **primary key** (*id*).

Exercici 2

Utilitzant JOIN realitzaràs les següents consultes:

- Llistat dels països que estan generant vendes.

```
USE transactions;
```

- ```
SELECT DISTINCT c.country AS paisos_transaccions
FROM company c
JOIN transaction t
ON c.id = t.company_id
WHERE t.declined != 1;
```

| paisos_transaccions |
|---------------------|
| Germany             |
| Australia           |
| United States       |
| New Zealand         |
| Norway              |
| United Kingdom      |
| Italy               |
| Belgium             |
| Sweden              |
| Ireland             |
| China               |
| Canada              |
| France              |

| #      | Time     | Action                                                                                                                          | Message                |
|--------|----------|---------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 100108 | 16:59:05 | SELECT * FROM transactions.company                                                                                              | 100 row(s) returned    |
| 100109 | 16:59:06 | SELECT * FROM transactions.transaction                                                                                          | 100000 row(s) returned |
| 100110 | 17:00:39 | SELECT DISTINCT c.country FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined != 1                        | 15 row(s) returned     |
| 100111 | 17:09:51 | SELECT DISTINCT c.country AS paisos_transaccions FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined != 1 | 15 row(s) returned     |

He fet un JOIN de les dues taules per filtrar les transaccions que no són rebutjades i seleccionar els distints països.

- Des de quants països es generen les vendes.

```
SELECT COUNT(DISTINCT c.country) AS quantitat_paisos
FROM company c
JOIN transaction t
ON c.id = t.company_id
WHERE t.declined != 1;
```

| quantitat_paisos |
|------------------|
| 15               |

Result 4 x

Output

Action Output

| #      | Time     | Action                                                                                                                              | Message            |
|--------|----------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 100110 | 17:00:39 | SELECT DISTINCT c.country FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined != 1                            | 15 row(s) returned |
| 100111 | 17:09:51 | SELECT DISTINCT c.country AS paisos_transaccions FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined != 1     | 15 row(s) returned |
| 100112 | 17:18:53 | SELECT COUNT(DISTINCT c.country) FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined != 0                     | 1 row(s) returned  |
| 100113 | 17:19:55 | SELECT COUNT(DISTINCT c.country) AS quantitat_paisos FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined != 0 | 1 row(s) returned  |

Aquí el mateix que abans, posant el filtre de les rebutjades, però afegint el *COUNT* per contar els diferents països.

- Identifica la companyia amb la mitjana més gran de vendes.

```
SELECT c.company_name AS nom, c.id AS id_companyia, ROUND(AVG(t.amount), 2) AS mitjana_vendes
FROM transaction t
JOIN company c
ON t.company_id = c.id
WHERE t.declined != 1
GROUP BY c.company_name, c.id
ORDER BY mitjana_vendes DESC
LIMIT 1;
```

|   | nom                       | id_companyia | mitjana_vendes |
|---|---------------------------|--------------|----------------|
| ▶ | Ac Fermentum Incorporated | b-2222       | 284.91         |

| #      | Time     | Action                                                                                                                                   | Message            |
|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 100111 | 17:09:51 | SELECT DISTINCT c.country AS paisos_transaccions FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined != 1          | 15 row(s) returned |
| 100112 | 17:18:53 | SELECT COUNT(DISTINCT c.country) FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined != 0                          | 1 row(s) returned  |
| 100113 | 17:19:55 | SELECT COUNT(DISTINCT c.country) AS quantitat_paisos FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined != 0      | 1 row(s) returned  |
| 100114 | 17:29:46 | SELECT c.company_name AS nom, c.id AS id_companyia, ROUND(AVG(t.amount), 2) AS mitjana_vendes FROM transaction t JOIN company c ON t.... | 1 row(s) returned  |

Aquí he fet el mateix *JOIN*, he agrupat les companyies per calcular la mitjana amb *AVG*.

### Exercici 3

Utilitzant només subconsultes (sense utilitzar *JOIN*):

- Mostra totes les transaccions realitzades per empreses d'Alemanya.

```
1 • SELECT *
2 FROM transaction
3 WHERE company_id IN (
4 SELECT id
5 FROM company
6 WHERE country = "Germany") AND declined != 1;
```

|   | id                                   | credit_card_id | company_id | user_id | lat     | longitude | timestamp           | amount | declined |
|---|--------------------------------------|----------------|------------|---------|---------|-----------|---------------------|--------|----------|
| ▶ | 00138D38-206D-4C03-94B7-63A2676EB984 | Cc5-4899       | b-2222     | 318     | 41.3781 | 12.447    | 2020-03-25 10:43:43 | 426.36 | 0        |
|   | 0013C1B6-3884-4D6C-8154-E2B3FEBCA8E9 | Cc5-5070       | b-2222     | 489     | 41.3814 | 2.18176   | 2020-12-17 18:15:37 | 316.90 | 0        |
|   | 00201A11-2E62-44C4-941D-198FC8D877F0 | CcU-3512       | b-2222     | 193     | 55.5704 | -3.65129  | 2021-01-22 23:44:27 | 453.04 | 0        |
|   | 00235618-0A5C-4D49-9DCB-83A9405D8923 | Cc5-8137       | b-2222     | 3556    | 59.8421 | 18.729    | 2020-09-09 15:43:19 | 263.14 | 0        |
|   | 005A5A7B-1F1A-4B6C-9B15-1625A78C9C38 | Cc5-8998       | b-2222     | 4417    | 41.1591 | -8.63905  | 2024-05-15 09:10:11 | 442.01 | 0        |
|   | 00687139-4882-4FFA-8E73-820376F04A84 | Cc5-4870       | b-2222     | 289     | 51.1966 | 10.4669   | 2019-03-09 19:37:49 | 524.84 | 0        |
|   | 0074F4DD-32F1-4827-8758-55896314623A | Cc5-8081       | b-2222     | 3500    | 39.7016 | -8.50325  | 2016-12-26 23:06:57 | 491.90 | 0        |
|   | 00AAB9CD-39D6-4DCB-8A1D-138E73DC90A9 | Cc5-6797       | b-2222     | 2216    | 55.7652 | -3.76245  | 2021-04-25 03:06:59 | 167.15 | 0        |
|   | 00BE9D04-6920-47D8-ABE8-325E2269829D | Cc5-4983       | b-2222     | 402     | 38.708  | -9.12993  | 2019-02-27 15:25:16 | 141.66 | 0        |
|   | 00DA0383-E048-4577-8ED1-3C56C258FF2F | Cc5-9223       | b-2222     | 4642    | 51.1742 | 10.2027   | 2019-03-21 11:47:34 | 325.62 | 0        |
|   | 00DD11DE-ED01-4BBD-93A0-174D183A59DF | Cc5-7681       | b-2222     | 3100    | 45.7565 | 4.83109   | 2024-01-28 18:20:49 | 242.53 | 0        |
|   | 01449CE0-98E9-4DE5-9810-728C68A00E6F | Cc5-5424       | b-2222     | 843     | 47.0163 | 2.26064   | 2024-02-17 19:37:14 | 451.71 | 0        |
|   | 0176F8F7-741F-47D4-ARR9-9F246D9F472F | Cc5-7510       | b-2222     | 2929    | 52.0619 | 4.79464   | 2021-08-28 16:29:38 | 9.46   | 0        |

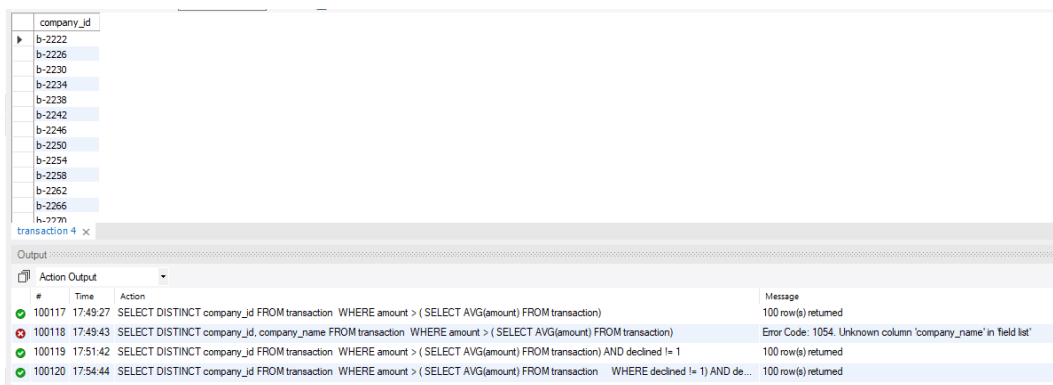
  

| #      | Time     | Action                                                                                                                                   | Message                                              |
|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 100113 | 17:19:55 | SELECT COUNT(DISTINCT c.country) AS quantitat_paisos FROM company c JOIN transaction t ON c.id = t.company_id WHERE t.declined != 0      | 1 row(s) returned                                    |
| 100114 | 17:29:46 | SELECT c.company_name AS nom, c.id AS id_companyia, ROUND(AVG(t.amount), 2) AS mitjana_vendes FROM transaction t JOIN company c ON t.... | 1 row(s) returned                                    |
| 100115 | 17:39:26 | SELECT * FROM transaction WHERE company_id IN ( SELECT id, company_name FROM company WHERE country = "Germany") AND declined != 1        | Error Code: 1241. Operand should contain 1 column(s) |
| 100116 | 17:40:04 | SELECT * FROM transaction WHERE company_id IN ( SELECT id FROM company WHERE country = "Germany") AND declined != 1                      | 13269 row(s) returned                                |

He fet una subconsulta on hi ha una llista d'empreses d'Alemania, i he usat */N* per seleccionar les transaccions on coincideix l'empresa.

- Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```
8 • SELECT DISTINCT company_id
9 FROM transaction
10 WHERE amount > (
11 SELECT AVG(amount)
12 FROM transaction
13 WHERE declined != 1) AND declined != 1;
```

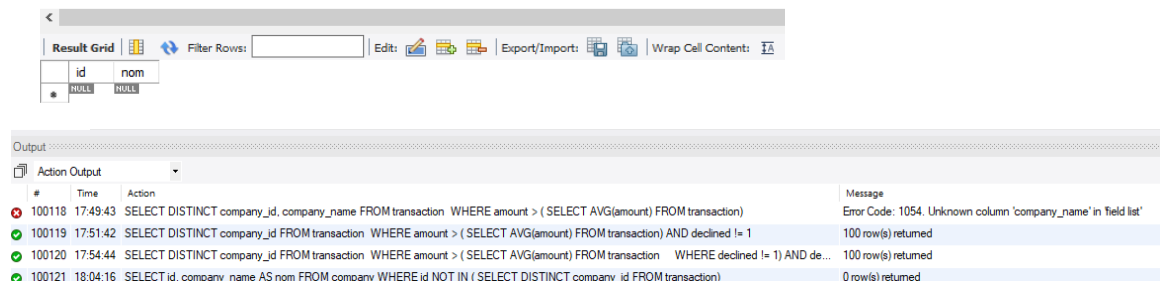


The screenshot shows a database interface. On the left, there is a list of company IDs: b-2222, b-2226, b-2230, b-2234, b-2238, b-2242, b-2246, b-2250, b-2254, b-2258, b-2262, b-2266, and h-2270. Below this list, there is a tab labeled 'transaction 4'. The main area shows the output of the SQL query, which is a table with columns: #, Time, Action, and Message. The output log shows four entries, with the last one indicating that 100 rows were returned for the query.

He fet una subconsulta calculant la mitjana total de la columna *amount* per poder filtrar les que són superiors a aquesta mitjana.

- Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

```
15 • SELECT id, company_name AS nom
16 FROM company
17 WHERE id NOT IN (
18 SELECT DISTINCT company_id
19 FROM transaction);
```



The screenshot shows a database interface. At the top, there is a 'Result Grid' with columns 'id' and 'nom'. Below it, there is a tab labeled 'transaction 4'. The main area shows the output of the SQL query, which is a table with columns: #, Time, Action, and Message. The output log shows four entries, with the last one indicating that 0 rows were returned for the query.

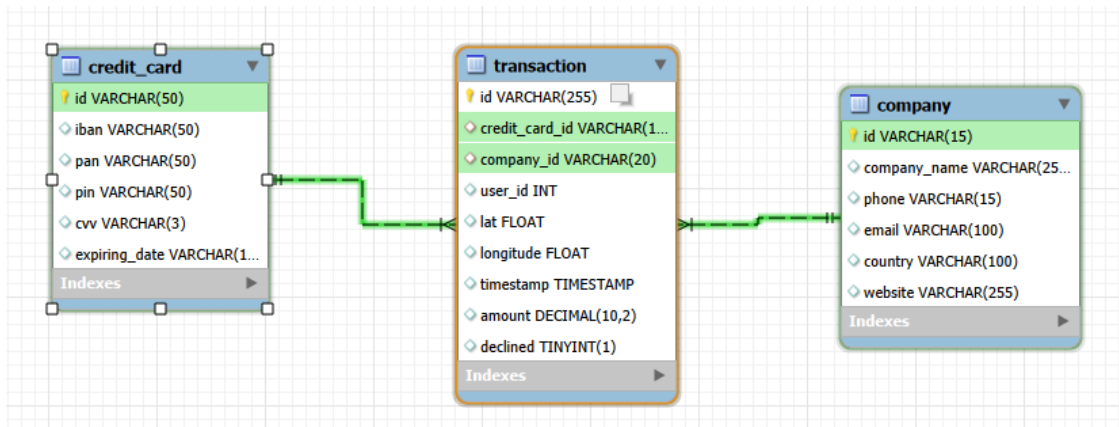
He fet una subconsulta per seleccionar les empreses que no surten a la llista *transactions*.

## Exercici 4

La teva tasca és dissenyar i crear una taula anomenada "credit\_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades\_introduir\_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest

```
1 • CREATE TABLE credit_card (
2 id VARCHAR(50),
3 iban VARCHAR(50),
4 pan VARCHAR(50),
5 pin VARCHAR(50),
6 cvv VARCHAR(3),
7 expiring_date VARCHAR(10)
8);
9
10 • ALTER TABLE credit_card
11 ADD PRIMARY KEY (id);
12
13 • ALTER TABLE transaction
14 ADD CONSTRAINT fk_transaction_card
15 FOREIGN KEY (credit_card_id)
16 REFERENCES credit_card(id);
```

| Action Output |          |                                                                                                                                                  |  |                                                                  |
|---------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------|
| #             | Time     | Action                                                                                                                                           |  | Message                                                          |
| ✓ 105126      | 18:15:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9580', 'XX781258889851950806677358', '5541182364498931', '9273'... |  | 1 row(s) affected                                                |
| ✓ 105127      | 18:15:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9581', 'XX915670516405388124398147', '2624305470167630', '4336'... |  | 1 row(s) affected                                                |
| ✓ 105128      | 18:31:51 | ALTER TABLE credit_card ADD PRIMARY KEY (id)                                                                                                     |  | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0           |
| ✓ 105129      | 18:31:54 | ALTER TABLE transaction ADD CONSTRAINT fk_transaction_card FOREIGN KEY (credit_card_id) REFERENCES credit_card(id)                               |  | 100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0 |



La taula *credit\_card* té relació directa amb la taula *transaction*. La relació es 1 (*credit\_card*) a N (*transaction*). La seva **primary key** és *id*, que és **foreign key** a *transaction* a la columna *credit\_card\_id*.

## Exercici 5

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.

```
1 • UPDATE credit_card
2 SET iban = "TR323456312213576817699999"
3 WHERE id = "CcU-2938";
```

| Output        |          |                                                                                                                                                  |                                                                  |
|---------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Action Output |          |                                                                                                                                                  |                                                                  |
| #             | Time     | Action                                                                                                                                           | Message                                                          |
| 105127        | 18:15:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-9581', 'XX915670516405388124398147', '2624305470167630', '4336'... | 1 row(s) affected                                                |
| 105128        | 18:31:51 | ALTER TABLE credit_card ADD PRIMARY KEY (id)                                                                                                     | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0           |
| 105129        | 18:31:54 | ALTER TABLE transaction ADD CONSTRAINT fk_transaction_card FOREIGN KEY (credit_card_id) REFERENCES credit_card(id)                               | 100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0 |
| 105130        | 18:40:53 | UPDATE credit_card SET iban = "TR323456312213576817699999" WHERE id = "CcU-2938"                                                                 | 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0         |

|   | id       | iban                       | pan              | pin  | cvv  | expiring_date |
|---|----------|----------------------------|------------------|------|------|---------------|
| ▶ | CcU-2938 | TR323456312213576817699... | 5424465566813633 | 3257 | 984  | 10/30/22      |
| ✱ | NULL     | NULL                       | NULL             | NULL | NULL | NULL          |

## Exercici 6

En la taula "transaction" ingressa una nova transacció amb la següent informació:

- `INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date)`  
`VALUES ("CcU-9999", "TR323456312213576817699888", "5424465566813622", "1234", "123", "11/30/22");`
- `INSERT INTO company (id, company_name, phone, email, country, website)`  
`VALUES ("b-9999", "Jamon SL", "01 02 03 04 05", "jamon@sl.com", "Andorra", "https://jamon.sl");`
- `INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)`  
`VALUES ("108B1D1D-5B23-A76C-55EF-C568E49A99DD", "CcU-9999", "b-9999", "9999", 829.999, -117.999, 111.11, 0);`

| Action Output |          |                                                                                                                                                |                                                                   |
|---------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| #             | Time     | Action                                                                                                                                         | Message                                                           |
| 105132        | 19:08:56 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ("CcU-9999", "TR323456312213576817699888", "5424465566813622", ...     | 1 row(s) affected                                                 |
| 105133        | 19:08:59 | INSERT INTO company (id, company_name, phone, email, country, website) VALUES ("b-9999", "Jamon SL", "01 02 03 04 05", "jamon@sl.com", "...    | Error Code: 1136. Column count doesn't match value count at row 1 |
| 105134        | 19:09:44 | INSERT INTO company (id, company_name, phone, email, country, website) VALUES ("b-9999", "Jamon SL", "01 02 03 04 05", "jamon@sl.com", "...    | 1 row(s) affected                                                 |
| 105135        | 19:09:47 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ("108B1D1D-5B23-A76C-55EF-C568E4... | 1 row(s) affected                                                 |

Com a la taula *transaction* hi ha dades que són **foreign key**, primer s'han d'afegir a les respectives taules que són **primary key**. Les dades de *credit\_card* i *company* en aquest cas són inventades.

## Exercici 7

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit\_card. Recordar mostrar el canvi realitzat.

```
1 • ALTER TABLE credit_card
2 DROP pan;
3
4 • SELECT *
5 FROM credit_card;
```

|   | id       | iban                     | pin  | cvv | expiring_date |
|---|----------|--------------------------|------|-----|---------------|
| ▶ | Cc5-4857 | XX4857591835292505850771 | 1819 | 467 | 09/27/25      |
|   | Cc5-4858 | XX8581768137002436094025 | 3964 | 817 | 12/28/28      |
|   | Cc5-4859 | XX7826930491423553609370 | 4983 | 277 | 11/26/26      |
|   | Cc5-4860 | XX555990368835304645299  | 6876 | 661 | 07/27/27      |
|   | Cc5-4861 | XX2035182877195191627307 | 5710 | 398 | 04/25/26      |
|   | Cc5-4862 | XX4774721462463645409758 | 4042 | 174 | 11/27/26      |
|   | Cc5-4863 | XX1476829664245046207111 | 5969 | 449 | 12/27/29      |
|   | Cc5-4864 | XX8380298893385731196159 | 8481 | 139 | 02/28/26      |
|   | Cc5-4865 | XX7085078596101025280599 | 7847 | 903 | 11/25/28      |
|   | Cc5-4866 | XX4792859188206596406839 | 9271 | 961 | 02/28/25      |
|   | Cc5-4867 | XX6036298816319374853717 | 4820 | 862 | 11/30/25      |
|   | Cc5-4868 | XX3929101617300533068044 | 6860 | 549 | 08/25/25      |
| ▼ | Cc5-4869 | XX1566712086111531886485 | 4775 | 448 | 09/29/25      |

credit\_card 1 x

Output

| #      | Time     | Action                                                                                                                                         | Message                                                |
|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 105134 | 19:09:44 | INSERT INTO company (id, company_name, phone, email, country, website) VALUES ('b-9999', 'Jamon SL', '01 02 03 04 05', 'jamon@sl.com', '...    | 1 row(s) affected                                      |
| 105135 | 19:09:47 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ('108B1D1D-5B23-A76C-55EF-C568E4... | 1 row(s) affected                                      |
| 105136 | 19:25:07 | ALTER TABLE credit_card DROP pan                                                                                                               | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |
| 105137 | 19:25:10 | SELECT * FROM credit_card                                                                                                                      | 5001 row(s) returned                                   |

## Exercici 8

Descarrega els arxius CSV que trobaràs a l'apartat de recursos:

- american\_users.csv
- european\_users.csv
- companies.csv
- credit\_cards.csv
- transactions.csv

Estudia'ls i dissenya una base de dades amb un esquema d'estrella que contingui, almenys 4 taules de les quals puguis realitzar les següents consultes:

La taula de products.csv l'utilitzarem més endavant.

```
• CREATE DATABASE ventes;
• USE ventes;
```

```
• CREATE TABLE transactions (
 id VARCHAR(50),
 card_id VARCHAR(50),
 business_id VARCHAR(50),
 timestamp VARCHAR(50),
 amount FLOAT,
 declined VARCHAR(1),
 product_ids VARCHAR(50),
 user_id VARCHAR(50),
 lat FLOAT,
 longitude FLOAT,
 discount_amount FLOAT,
 tax_amount FLOAT,
 shipping_amount FLOAT,
 channel VARCHAR(50),
 campaign_id VARCHAR(50),
 device_type VARCHAR(50),
 is_international VARCHAR(1),
 decline_reason VARCHAR(50),
 distance_km FLOAT);
```

```
• CREATE TABLE targeta (
 id VARCHAR(50),
 user_id VARCHAR(50),
 iban VARCHAR(50),
 pan VARCHAR(50),
 pin VARCHAR(50),
 cvv VARCHAR(50),
 track1 VARCHAR(50),
 track2 VARCHAR(50),
 expiring_date VARCHAR(50),
 card_type VARCHAR(50),
 card_renewal_flag VARCHAR(50));
```

```
• CREATE TABLE usuari (
 id VARCHAR(50),
 name VARCHAR(50),
 surname VARCHAR(50),
 phone VARCHAR(50),
 email VARCHAR(50),
 birth_date VARCHAR(50),
 country VARCHAR(50),
 city VARCHAR(50),
 postal_code VARCHAR(50),
 adress VARCHAR(50),
 signup_date VARCHAR(50),
 user_segment VARCHAR(50),
 income_band VARCHAR(50));
```

```

i3 • CREATE TABLE empreses (
i4 company_id VARCHAR(50),
i5 company_name VARCHAR(50),
i6 phone VARCHAR(50),
i7 email VARCHAR(50),
i8 country VARCHAR(50),
i9 website VARCHAR(50),
i0 merchan_category VARCHAR(50),
i1 merchan_price_position VARCHAR(50));
i2
• CREATE TABLE productes (
 id VARCHAR(50),
 product_name VARCHAR(50),
 price FLOAT,
 colour VARCHAR(50),
 weight FLOAT,
 warehouse_id VARCHAR(50),
 category VARCHAR(50),
 brand VARCHAR(50),
 cost FLOAT,
 launch_date VARCHAR(20));

```

```

mysql> LOAD DATA LOCAL INFILE 'C:/Users/adria/Downloads/N1-Ex.8_transactions.csv'
-> INTO TABLE transactions
-> FIELDS TERMINATED BY ';'
-> ENCLOSED BY '"'
-> LINES TERMINATED BY '\n'
-> IGNORE 1 LINES;
Query OK, 100000 rows affected (2.24 sec)
Records: 100000 Deleted: 0 Skipped: 0 Warnings: 0

mysql> LOAD DATA LOCAL INFILE "C:/Users/adria/Downloads/N1-Ex.8_credit_cards.csv"
-> INTO TABLE targeta
-> FIELDS TERMINATED BY ','
-> ENCLOSED BY '"'
-> LINES TERMINATED BY '\n'
-> IGNORE 1 LINES;
Query OK, 5000 rows affected, 4725 warnings (0.17 sec)
Records: 5000 Deleted: 0 Skipped: 0 Warnings: 4725

mysql> LOAD DATA LOCAL INFILE "C:/Users/adria/Downloads/N1-Ex.8_companies.csv"
-> INTO TABLE empreses
-> FIELDS TERMINATED BY ','
-> ENCLOSED BY '"'
-> LINES TERMINATED BY '\n'
-> IGNORE 1 LINES;
Query OK, 100 rows affected, 2 warnings (0.01 sec)
Records: 100 Deleted: 0 Skipped: 0 Warnings: 2

mysql> LOAD DATA LOCAL INFILE "C:/Users/adria/Downloads/N1-Ex.8_european_users.csv"
-> INTO TABLE usuarios
-> FIELDS TERMINATED BY ','
-> ENCLOSED BY '"'
-> LINES TERMINATED BY '\n'
-> IGNORE 1 LINES;
Query OK, 3990 rows affected (0.23 sec)
Records: 3990 Deleted: 0 Skipped: 0 Warnings: 0

mysql> LOAD DATA LOCAL INFILE "C:/Users/adria/Downloads/N1-Ex.8_american_users.csv"
-> INTO TABLE usuarios
-> FIELDS TERMINATED BY ','
-> ENCLOSED BY '"'
-> LINES TERMINATED BY '\n'
-> IGNORE 1 LINES;
Query OK, 1010 rows affected (0.03 sec)
Records: 1010 Deleted: 0 Skipped: 0 Warnings: 0

mysql> LOAD DATA LOCAL INFILE "C:/Users/adria/Downloads/N1-Ex.8_products.csv"
-> INTO TABLE productes
-> FIELDS TERMINATED BY ','
-> ENCLOSED BY '"'
-> LINES TERMINATED BY '\n'
-> IGNORE 1 LINES;
Query OK, 100 rows affected, 200 warnings (0.01 sec)
Records: 100 Deleted: 0 Skipped: 0 Warnings: 200

```

- ALTER TABLE empreses  
ADD PRIMARY KEY (company\_id);
- ALTER TABLE usuarios  
ADD PRIMARY KEY (id);
- ALTER TABLE targeta  
ADD PRIMARY KEY (id);
- ALTER TABLE transaccions  
ADD PRIMARY KEY (id),  
ADD CONSTRAINT fk\_trans\_targ  
FOREIGN KEY (card\_id)  
REFERENCES targeta(id),  
ADD CONSTRAINT fk\_trans\_empr  
FOREIGN KEY (business\_id)  
REFERENCES empreses(company\_id),  
ADD CONSTRAINT fk\_trans\_usua  
FOREIGN KEY (user\_id)  
REFERENCES usuarios(id);
- ALTER TABLE targeta  
ADD CONSTRAINT fk\_targ\_usua  
FOREIGN KEY (user\_id)  
REFERENCES usuarios(id);

Aquesta base de dades té un model d'estrella on la taula de fets és la taula *transaccions*, que la seva **primary key** és *id*, i les seves **foreign keys** són *user\_id* (**primary key** a *usuarios*), *business\_id* (**primary key** a *empreses*) i *card\_id* (**primary key** a *targeta*). Té relació n a 1 amb la resta de taules. També hi ha relació 1 (*usuarios*) a n (*targeta*), que la taula *targeta* té la **foreign key** *user\_id* (**primary key** a *usuarios*).



## Exercici 10

Mostra la mitjana d'amount per IBAN de les targetes de crèdit a la companyia Donec Ltd, utilitza almenys 2 taules.

```
1 • SELECT g.iban AS targeta_iban, ROUND(AVG(t.amount), 2) AS mitjana_trans
2 FROM transaccions t
3 JOIN targeta g
4 ON t.card_id = g.id
5 JOIN empreses e
6 ON t.business_id = e.company_id
7 WHERE e.company_name = "Donec Ltd"
8 GROUP BY g.iban;
```

He fet un triple JOIN amb transaccions, targeta i empreses. He filtrat per nom d'empreses i he agrupat per iban per poder calcular la mitjana.

| targeta_iban                | mitjana_trans |
|-----------------------------|---------------|
| CA103460001848465124357723  | 480.35        |
| CA107154170758509463327452  | 100.79        |
| CA10756236329629763054895   | 115.05        |
| CA133060727799428001951351  | 625.88        |
| CA134337007195142681679763  | 279.27        |
| CA137550525951024363769804  | 287.53        |
| CA138345948053324736749079  | 378.24        |
| CA138798042349017958158099  | 372.12        |
| CA139534239296395775858587  | 586.17        |
| CA144137745899914825332907  | 474.48        |
| CA149881306418169830767362  | 202.69        |
| CA151913509660987161349266  | 429.8         |
| CA1577366408015511156418160 | 160.74        |

Result 1 x

Output

| #      | Time     | Action                                                                                                                                                                                                                            | Message                                                          |
|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 105162 | 23:22:49 | ALTER TABLE transaccions ADD PRIMARY KEY (id), ADD CONSTRAINT fk_trans_jarg FOREIGN KEY (card_id) REFERENCES targeta(id), ADD C...                                                                                                | 100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0 |
| 105163 | 23:23:17 | ALTER TABLE targeta ADD CONSTRAINT fk_jarg_usua FOREIGN KEY (user_id) REFERENCES usuari(id)                                                                                                                                       | 5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0     |
| 105164 | 23:48:29 | SELECT * FROM usuari WHERE id IN ( SELECT user_id FROM transaccions WHERE declined = 0 GROUP BY user_id HAVING COUNT(id) > 80)                                                                                                    | 4 row(s) returned                                                |
| 105165 | 23:57:02 | SELECT g.iban AS targeta_iban, ROUND(AVG(t.amount), 2) AS mitjana_trans FROM transaccions t JOIN targeta g ON t.card_id = g.id JOIN empreses e ON t.business_id = e.company_id WHERE e.company_name = "Donec Ltd" GROUP BY g.iban | 371 row(s) returned                                              |

## NIVELL 2

### Exercici 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

```
SELECT DATE(timestamp) AS data, ROUND(SUM(amount), 2) AS total_vendes
FROM transaccions
WHERE declined = 0
GROUP BY data
ORDER BY total_vendes DESC
LIMIT 5;
```

A la taula transaccions agrupo per data per poder sumar. Després ordeno i faig el límit.

| data       | total_vendes |
|------------|--------------|
| 2023-02-20 | 15728.4      |
| 2019-11-18 | 15447.89     |
| 2022-12-13 | 14944        |
| 2019-03-18 | 14896.47     |
| 2017-12-20 | 14367.96     |

Result 1 x

Output

| #      | Time     | Action                                                                                                                                                                                                                            | Message                                                      |
|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 105163 | 23:23:17 | ALTER TABLE targeta ADD CONSTRAINT fk_jarg_usua FOREIGN KEY (user_id) REFERENCES usuari(id)                                                                                                                                       | 5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0 |
| 105164 | 23:48:29 | SELECT * FROM usuari WHERE id IN ( SELECT user_id FROM transaccions WHERE declined = 0 GROUP BY user_id HAVING COUNT(id) > 80)                                                                                                    | 4 row(s) returned                                            |
| 105165 | 23:57:02 | SELECT g.iban AS targeta_iban, ROUND(AVG(t.amount), 2) AS mitjana_trans FROM transaccions t JOIN targeta g ON t.card_id = g.id JOIN empreses e ON t.business_id = e.company_id WHERE e.company_name = "Donec Ltd" GROUP BY g.iban | 371 row(s) returned                                          |
| 105166 | 00:05:27 | SELECT DATE(timestamp) AS data, ROUND(SUM(amount), 2) AS total_vendes FROM transaccions WHERE declined = 0 GROUP BY data ORDER BY total_vendes DESC LIMIT 5                                                                       | 5 row(s) returned                                            |

## Exercici 2

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

```
1 • SELECT company_name AS nom, phone AS telefon, country AS pais, timestamp AS data, amount AS diners
2 FROM transaccions t
3 JOIN empreses e
4 ON t.business_id = e.company_id
5 WHERE t.amount BETWEEN 350 AND 400 AND DATE(timestamp) IN ("2015-04-29", "2018-07-20", "2024-03-13") AND t.declined = 0
6 ORDER BY amount DESC;
```

|   | nom                    | telefon          | pais          | data                | diners |
|---|------------------------|------------------|---------------|---------------------|--------|
| ▶ | Fringilla LLC          | +1 150 784 1716  | New Zealand   | 2015-04-29 06:18:25 | 390.33 |
|   | Auctor Mauris Vel LLP  | +32 133 447 8239 | United States | 2024-03-13 01:23:09 | 383.58 |
|   | Mauris Institute       | +75 473 619 9472 | Sweden        | 2015-04-29 21:24:49 | 369.2  |
|   | Pede Cum Ltd           | +26 625 798 6126 | Norway        | 2018-07-20 14:57:32 | 358.64 |
|   | Netus Et Malesuada Ltd | +52 818 402 1638 | Netherlands   | 2024-03-13 13:20:45 | 352.21 |
|   | Tempor Diam Institute  | +89 848 141 3136 | Netherlands   | 2024-03-13 20:49:51 | 350.71 |

Result 1 x

Output

| #      | Time     | Action                                                                                                                                                                 | Message             |
|--------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 105164 | 23:48:29 | SELECT * FROM usuarios WHERE id IN ( SELECT user_id FROM transaccions WHERE declined = 0 GROUP BY user_id HAVING COUNT(id) > 80)                                       | 4 row(s) returned   |
| 105165 | 23:57:02 | SELECT g.iban AS targeta_iban, ROUND(AVG(t.amount), 2) AS mitjana_trans FROM transaccions t JOIN targeta g ON t.card_id = g.id JOIN empreses e ON t.business_id = e.id | 371 row(s) returned |
| 105166 | 00:05:27 | SELECT DATE(timestamp) AS data, ROUND(SUM(amount), 2) AS total_ventas FROM transaccions WHERE declined = 0 GROUP BY data ORDER BY data                                 | 5 row(s) returned   |
| 105167 | 00:13:17 | SELECT company_name AS nom, phone AS telefon, country AS pais, timestamp AS data, amount AS diners FROM transaccions t JOIN empreses e ON t.business_id = e.id         | 6 row(s) returned   |

Faig un JOIN d'empreses i transaccions i poso els filtres que em demana.

## Exercici 3

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen igual o més de 400 transaccions o menys.

```
1 • SELECT e.company_name AS nom, t.quantitat_transaccions,
2 CASE
3 WHEN quantitat_transaccions >= 400 THEN "Moltes transaccions"
4 ELSE "Poques transaccions"
5 END AS activitat
6 FROM empreses e
7 JOIN (
8 SELECT business_id, COUNT(id) AS quantitat_transaccions
9 FROM transaccions
10 WHERE declined = 0
11 GROUP BY business_id) t
12 ON e.company_id = t.business_id;
```

|   | nom                          | quantitat_transaccions | activitat           |
|---|------------------------------|------------------------|---------------------|
| ▶ | Ac Fermentum Incorporated    | 2387                   | Moltes transaccions |
|   | Magna A Neque Industries     | 408                    | Moltes transaccions |
|   | Fusce Corp.                  | 444                    | Moltes transaccions |
|   | Convalis In Incorporated     | 1501                   | Moltes transaccions |
|   | Ante Iaculis Nec Foundation  | 471                    | Moltes transaccions |
|   | Donec Ltd                    | 443                    | Moltes transaccions |
|   | Sed Nunc Ltd                 | 1526                   | Moltes transaccions |
|   | Amet Nulla Donec Corporation | 1501                   | Moltes transaccions |
|   | Nascetur Ridiculus Mus Inc.  | 417                    | Moltes transaccions |
|   | Vestibulum Lorem PC          | 442                    | Moltes transaccions |
|   | Gravida Sagittis LLP         | 1467                   | Moltes transaccions |
|   | Mus Aenean Eget Foundation   | 1547                   | Moltes transaccions |
|   | The Pariant Teethtite        | 435                    | Moltes transaccions |

Result 1 x

Output

Action Output

| #      | Time     | Action                                                                                                                                      | Message             |
|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 105166 | 00:05:27 | SELECT DATE(timestamp) AS data, ROUND(SUM(amount), 2) AS total_ventas FROM transaccions WHERE declined = 0 GROUP BY data ORDER ...          | 5 row(s) returned   |
| 105167 | 00:13:17 | SELECT company_name AS nom, phone AS telefon, country AS pais, timestamp AS data, amount AS diners FROM transaccions t JOIN empreses e O... | 6 row(s) returned   |
| 105168 | 00:19:46 | SELECT e.company_name AS nom, t.quantitat_transaccions, CASE WHEN quantitat_transaccions >= 400 THEN "Moltes transaccions" ELSE "Po...      | 100 row(s) returned |
| 105169 | 00:20:03 | SELECT e.company_name AS nom, t.quantitat_transaccions, CASE WHEN quantitat_transaccions >= 400 THEN "Moltes transaccions" ELSE "Po...      | 100 row(s) returned |

He fet una subconsulta per fer una taula de transaccions per companyia. L'he unit amb empreses i he classificat per quantitat de transaccions.

## Exercici 4

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

• **DELETE FROM transaccions**

**WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD";**

Action Output

| #      | Time     | Action                                                                                                                                      | Message             |
|--------|----------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 105167 | 00:13:17 | SELECT company_name AS nom, phone AS telefon, country AS pais, timestamp AS data, amount AS diners FROM transaccions t JOIN empreses e O... | 6 row(s) returned   |
| 105168 | 00:19:46 | SELECT e.company_name AS nom, t.quantitat_transaccions, CASE WHEN quantitat_transaccions >= 400 THEN "Moltes transaccions" ELSE "Po...      | 100 row(s) returned |
| 105169 | 00:20:03 | SELECT e.company_name AS nom, t.quantitat_transaccions, CASE WHEN quantitat_transaccions >= 400 THEN "Moltes transaccions" ELSE "Po...      | 100 row(s) returned |
| 105170 | 00:25:38 | DELETE FROM transaccions WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD"                                                                  | 1 row(s) affected   |

## Exercici 5

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.

```

1 • CREATE OR REPLACE VIEW VistaMarketing AS
2 SELECT t. business_id AS id_comp, e.company_name AS nom, e.phone AS telefon, e.country AS pais, ROUND(AVG(t.amount), 2) AS mitjana
3 FROM empreses e
4 JOIN transaccions t
5 ON e.company_id = t.business_id
6 WHERE t.declined = 0
7 GROUP BY id_comp, nom, telefon, pais;
8
9 • SELECT *
10 FROM VistaMarketing
11 ORDER BY mitjana DESC;

```

| Result Grid        |                                   |                  |                |         |  |
|--------------------|-----------------------------------|------------------|----------------|---------|--|
| Filter Rows:       |                                   |                  |                |         |  |
| Export:            |                                   |                  |                |         |  |
| Wrap Cell Content: |                                   |                  |                |         |  |
| id_comp            | nom                               | telefon          | pais           | mitjana |  |
| b-2222             | Ac Fermentum Incorporated         | +64 601 244 6320 | Germany        | 330.71  |  |
| b-2346             | Ut Semper Foundation              | +10 283 500 6449 | Sweden         | 307.3   |  |
| b-2502             | Dolor Vitae Limited               | +6 421 567 9195  | France         | 306.21  |  |
| b-2574             | Orci Adipiscing Limited           | +60 897 390 5820 | United Kingdom | 305.87  |  |
| b-2542             | Nunc In Foundation                | +13 267 741 2619 | Italy          | 305.67  |  |
| b-2510             | Maecenas Malesuada Fringilla Inc. | +51 673 540 5289 | Netherlands    | 304.49  |  |
| b-2450             | Turpis Company                    | +17 624 311 7168 | Netherlands    | 303.47  |  |
| b-2558             | Nulla Integer Vulputate Corp.     | +75 236 687 9162 | Sweden         | 303.28  |  |
| b-2370             | Non Justo Corp.                   | +67 963 294 1666 | Sweden         | 303.17  |  |
| b-2518             | Aliquet Sem Limited               | +44 744 425 9915 | Netherlands    | 302.77  |  |
| b-2442             | Aliquam Erat Volutpat LLP         | +39 372 330 7229 | Italy          | 302.24  |  |
| b-2602             | Placerat LLP                      | +17 324 602 5033 | Netherlands    | 302.19  |  |
| b-7466             | Amet Lorem LLP                    | +40 468 847 7738 | Spain          | 301.11  |  |

| Output        |          |                                                                                                                                        |                     |  |
|---------------|----------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------|--|
| Action Output |          |                                                                                                                                        |                     |  |
| #             | Time     | Action                                                                                                                                 | Message             |  |
| 105169        | 00:20:03 | SELECT e.company_name AS nom,t.quantitat_transaccions, CASE WHEN quantitat_transaccions >= 400 THEN "Moltes transaccions" ELSE "Po...  | 100 row(s) returned |  |
| 105170        | 00:25:38 | DELETE FROM transaccions WHERE id = "000447FE-B650-4DCF-85DE-C7ED0EE1CAAD"                                                             | 1 row(s) affected   |  |
| 105171        | 00:31:54 | CREATE OR REPLACE VIEW VistaMarketing AS SELECT t.business_id AS id_comp, e.company_name AS nom, e.phone AS telefon, e.country AS p... | 0 row(s) affected   |  |
| 105172        | 00:32:34 | SELECT * FROM VistaMarketing ORDER BY mitjana DESC                                                                                     | 100 row(s) returned |  |

He creat la vista fent JOIN de les dues taules i agrupant per poder calcular la mitjana d'amount.

## NIVELL 3

### Exercici 1

Crea una nova taula que reflecteixi l'estat de les targetes de crèdit basat en si les tres últimes transaccions han estat declinades aleshores és inactiu, si almenys una no és rebutjada aleshores és actiu. Partint d'aquesta taula respon: Quantes targetes estan actives?

|    |   |                                                   |  |
|----|---|---------------------------------------------------|--|
| 1  | • | CREATE TABLE IF NOT EXISTS estat_targetes AS      |  |
| 2  |   | SELECT card_id,                                   |  |
| 3  |   | CASE                                              |  |
| 4  |   | WHEN ult_tra = 3 THEN 'inactiu'                   |  |
| 5  |   | ELSE 'actiu'                                      |  |
| 6  |   | END AS estat                                      |  |
| 7  |   | FROM (                                            |  |
| 8  |   | SELECT card_id, SUM(declined) AS ult_tra          |  |
| 9  |   | FROM (                                            |  |
| 10 |   | SELECT card_id, timestamp, declined,              |  |
| 11 |   | ROW_NUMBER() OVER(                                |  |
| 12 |   | PARTITION BY card_id                              |  |
| 13 |   | ORDER BY timestamp DESC                           |  |
| 14 |   | ) AS ultimes                                      |  |
| 15 |   | FROM transaccions                                 |  |
| 16 |   | ) AS c                                            |  |
| 17 |   | WHERE ultimes < 4                                 |  |
| 18 |   | GROUP BY card_id                                  |  |
| 19 |   | ) AS b;                                           |  |
| 20 |   |                                                   |  |
| 21 | • | SELECT COUNT(estat) AS quantitat_targetes_actives |  |
| 22 |   | FROM estat_targetes                               |  |
| 23 |   | WHERE estat = 'actiu';                            |  |

| Result Grid                |  |  |  |  |
|----------------------------|--|--|--|--|
| Filter Rows:               |  |  |  |  |
| Export:                    |  |  |  |  |
| Wrap Cell Content:         |  |  |  |  |
| quantitat_targetes_actives |  |  |  |  |
| 5000                       |  |  |  |  |

| Result 1      |          |                                                                                                                                        |                                                              |  |
|---------------|----------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--|
| Output        |          |                                                                                                                                        |                                                              |  |
| Action Output |          |                                                                                                                                        |                                                              |  |
| #             | Time     | Action                                                                                                                                 | Message                                                      |  |
| 105171        | 00:31:54 | CREATE OR REPLACE VIEW VistaMarketing AS SELECT t.business_id AS id_comp, e.company_name AS nom, e.phone AS telefon, e.country AS p... | 0 row(s) affected                                            |  |
| 105172        | 00:32:34 | SELECT * FROM VistaMarketing ORDER BY mitjana DESC                                                                                     | 100 row(s) returned                                          |  |
| 105173        | 00:39:22 | CREATE TABLE IF NOT EXISTS estat_targetes AS SELECT card_id, CASE WHEN ult_tra = 3 THEN 'inactiu' ELSE 'actiu' END AS e...             | 5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0 |  |
| 105174        | 00:40:31 | SELECT COUNT(estat) AS quantitat_targetes_actives FROM estat_targetes WHERE estat = 'actiu'                                            | 1 row(s) returned                                            |  |

Amb una window function he contat les targetes que hi ha agrupant les per data. Llavors he tret les que estan per sobre del 3 i les he classificat.

## Exercici 2

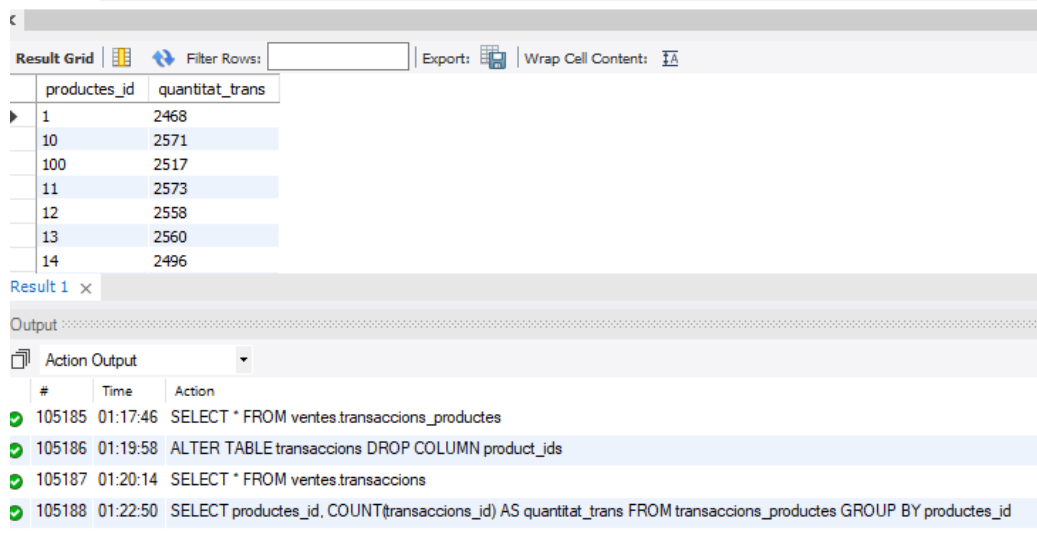
Crea una taula amb la qual puguem unir les dades de l'arxiu de products.csv amb la base de dades creada (ja que fins ara no podíem fer-ho), tenint en compte que des de transaction tens product\_ids. Genera la següent consulta:

```
1 • ALTER TABLE productes
2 MODIFY price DECIMAL(10, 2),
3 MODIFY cost DECIMAL(10, 2),
4 ADD PRIMARY KEY (id);
5
6 • CREATE TABLE IF NOT EXISTS transaccions_productes (
7 transaccions_id VARCHAR(50),
8 productes_id VARCHAR(50),
9 PRIMARY KEY (transaccions_id, productes_id),
10 FOREIGN KEY (transaccions_id) REFERENCES transaccions(id),
11 FOREIGN KEY (productes_id) REFERENCES productes(id));
12
13 • INSERT INTO transaccions_productes (transaccions_id, productes_id)
14 SELECT t.id AS transaccio, p.producte_id
15 FROM transaccions t,
16 JSON_TABLE(
17 CONCAT('[', t.product_ids, ']'),
18 '$[*]' COLUMNS (
19 producte_id VARCHAR(15) PATH '$') AS p;
20
21 • ALTER TABLE transaccions
22 DROP COLUMN product_ids;
```

Primer, posem la **primary key** a *productes*. Després he creat una taula pont entre *transaccions* i *productes* amb les seves característiques. Després he usat la funció JSON table per transformar *product\_ids* en una llista i poder posar dades a la taula. Per últim he eliminat *product\_ids* perquè ja no feia falta.

Necessitem conèixer el nombre de vegades que s'ha venut cada producte.

```
24 • SELECT productes_id, COUNT(transaccions_id) AS quantitat_trans
25 FROM transaccions_productes
26 GROUP BY productes_id;
```

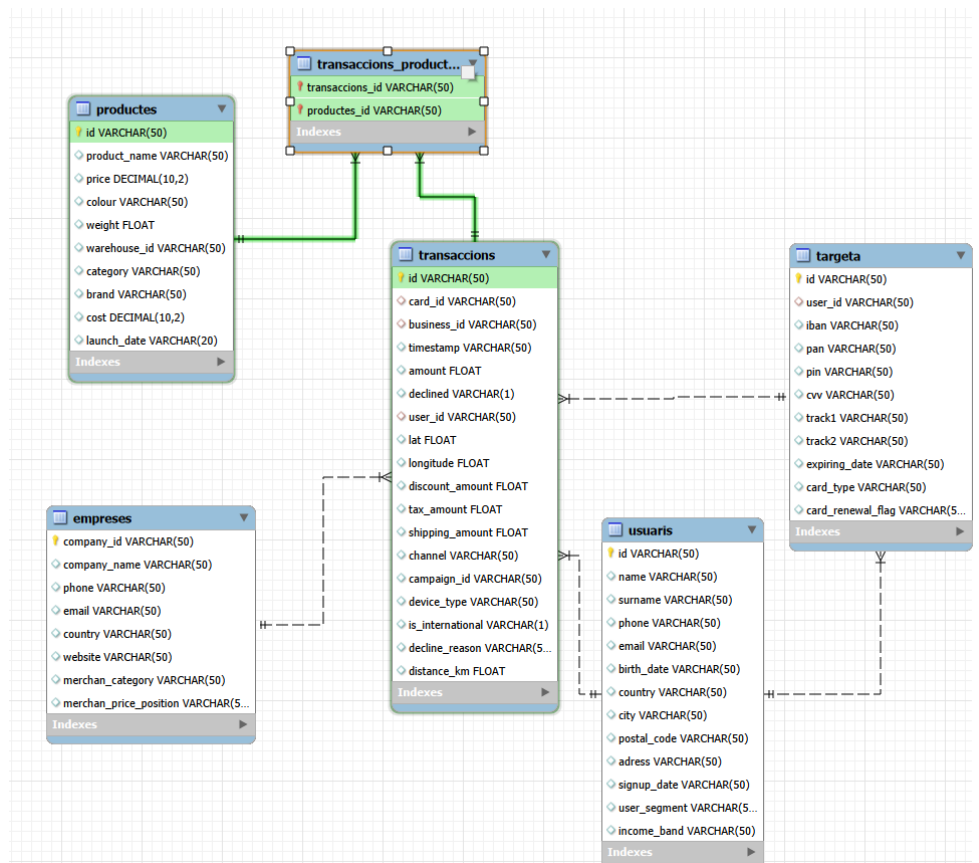


| productes_id | quantitat_trans |
|--------------|-----------------|
| 1            | 2468            |
| 10           | 2571            |
| 100          | 2517            |
| 11           | 2573            |
| 12           | 2558            |
| 13           | 2560            |
| 14           | 2496            |

| #      | Time     | Action                                                                                                           |
|--------|----------|------------------------------------------------------------------------------------------------------------------|
| 105185 | 01:17:46 | SELECT * FROM ventes.transaccions_productes                                                                      |
| 105186 | 01:19:58 | ALTER TABLE transaccions DROP COLUMN product_ids                                                                 |
| 105187 | 01:20:14 | SELECT * FROM ventes.transaccions                                                                                |
| 105188 | 01:22:50 | SELECT productes_id, COUNT(transaccions_id) AS quantitat_trans FROM transaccions_productes GROUP BY productes_id |

Aquí he agrupat els productes per contar les transaccions per producte.



La taula pont té una relació n a 1 amb *productes* i *transaccions*. Les dues columnes són **primary key** i a la vegada **foreign key** de les altres dues taules.